



Illuminating
ENGINEERING SOCIETY

TECHNICAL MEMORANDUM:
STANDARD FORMAT FOR THE
ELECTRONIC DATA TRANSFER OF
LIGHT OUTPUT MAINTENANCE
CHARACTERISTICS OF SOLID-STATE
LIGHT SOURCES

AN AMERICAN NATIONAL STANDARD



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Publication of this Technical Memorandum
has been approved by IES.
Suggestions for revisions
should be directed to IES.

Prepared by the IES Computer Committee



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1.0 Introduction and Scope

1.1 Introduction

LM-80 tests are conducted to measure the light output maintenance characteristics of solid-state light sources. The resulting data typically targets two commercial uses: 1) reports formatted for human reading (e.g., an LM-80 [see **Section 2.2**] report); or 2) LED lifetime projection calculators (e.g., ANSI/IES TM-21¹).

Data collected during an LM-80 maintenance test may have other uses beyond these immediate commercial applications. This is particularly true if multiple diverse LM-80 datasets are considered. For example, with multiple datasets, a researcher could use a review-based approach to investigate the correlation between a widely used packaging technology and a particular output maintenance characteristic. By including datasets from multiple manufacturers, trends that relate to the underlying packaging technology, rather than a specific commercial implementation, might be revealed. The revealed trends could then be used to focus subsequent primary research that would seek the root cause. The resulting improved understanding of the packaging technology could then stimulate improvements that could be applied to future products.

Most labs do save LM-80 data; however, data is collected using equipment that varies from laboratory to laboratory, and the data is typically stored in proprietary databases. Because of this, it is not always easy to access or exchange raw LM-80 test data.

This document provides a standard format for the electronic transfer of LM-80 test data in XML format. All required LM-80 report data items are included, as well as optional data items such as the spectral power distribution (SPD). The ordering of data elements does not follow the LM-80 report format; rather, the ordering is more data-centric, with administrative elements first, followed by device-under-test (DUT) items, test conditions, and then results. Where important, the precision required for a particular data element is specified; this helps to reduce uncertainty when the data is used for calculations. The format is described first

in human-readable text, and then computer-readable XML and JSON schemas are provided. Finally, sample XML and JSON documents are included in **Annex A**, and an overview of the document structure is provided in **Annex B**, to aid in picturing the hierarchy.

1.2 Scope

This Technical Memorandum (TM) describes the storage and transfer file format for LM-80 maintenance test data for solid-state light sources, such as LED packages, arrays, and modules; or laser diode packages, arrays, and modules.

This TM does not cover the storage and transfer of derivative data, such as TM-21 projection data.

2.0 Normative References

2.1 ECMA-404

ECMA International. The JSON Data Interchange Syntax, 2nd ed. Geneva: ECMA International; 2017. Online: <https://www.ecma-international.org/>. (Accessed 2023 Aug 31).

2.2 ANSI/IES LM-80-21

Illuminating Engineering Society. Approved Method: Measuring Maintenance of Light Output Characteristics of Solid-State Light Sources. New York: IES; 2021.

2.3 ANSI/IES TM-27-20

Illuminating Engineering Society. Technical Memorandum: IES Standard File Format for the Electronic Transfer of Spectral Data. New York: IES; 2020.

2.4 RFC 4122

Leach P, Mealling M, Salz R. A Universally Unique Identifier (UUID) URN Namespace. Internet Engineering Task Force; 2015 Jul. (RFC 4122). Online: <https://datatracker.ietf.org/doc/html/rfc4122>. (Accessed 2023 Aug 31).

2.5 W3C Extensible Markup Language (XML) 1.1, 2nd ed.

(<http://www.w3.org/TR/xml11>). (Accessed 2023 Aug 31).

2.6 W3C XML Schema Definition Language (XSD) 1.1. Part 1: Structures

(<http://www.w3.org/TR/xmlschema11-1>). (Accessed 2023 Aug 31).

2.7 W3C XML Schema Definition Language (XSD) 1.1. Part 2: Datatypes

(<http://www.w3.org/TR/xmlschema11-2>). (Accessed 2023 Aug 31).

3.0 Definitions

Definitions for the terms used in this TM can be found in normative reference ANSI/IES LM-80-21 (see **Section 2.2**).

4.0 XML Format for Light Output Maintenance Characteristics

The light output maintenance characteristics document shall follow the W3C XML Schema (see **Sections 2.6** and **2.7**) presented in **Section 5**. It shall include the required elements listed in this section, and it may include the optional elements listed in this section.

The XML document is separated into six elements:

- The required `Version` element (**Section 4.1**), which identifies the XML schema version.
- The required `Admin` element (**Section 4.2**), which contains administrative information.
- The required `DeviceUnderTest` element (**Section 4.3**), which contains information about the device under test.
- The required `TestEquipment` element (**Section 4.4**), which contains information about the environmental conditions and equipment used to perform the measurements.
- The required `MeasurementSet` element (**Section 4.5**), which contains information about test results.
- The optional `CustomData` element (**Section 4.6**), which contains information specific to a particular application; for example, government-mandated environmental criteria.

4.1 Version Element

The required `Version` element identifies the XML schema version. It shall be "1.0".

4.2 Admin Element

The required `Admin` element is the parent of the administrative information section. `Admin` elements are listed in **Table 4-1**.

4.2.1 Testing Agency Element. The required `TestAgency` element identifies the testing agency name that performed the data measurements.

Table 4-1. Admin Element Fields

Element Description	Element Name	Data Type	Required/Optional	Document Section
Testing Agency	TestAgency	xs:string	Required	4.2.1
Report Number	ReportNumber	xs:string	Required	4.2.2
Report Date	ReportDate	xs:date	Required	4.2.3
Test Start Date	TestStartDate	xs:date	Required	4.2.4
Test End Date	TestEndDate	xs:date	Required	4.2.5
Unique Identifier	UniqueIdentifier	xs:string	Required	4.2.6
Comment	Comment	xs:string	Optional	4.2.7
Reference	Reference	xs:string	Optional	4.2.8
More Information URI	MoreInfoURI	xs:anyURI	Optional	4.2.9

4.2.2 Report Number Element. The required `ReportNumber` element identifies the testing laboratory report number.

4.2.3 Report Date Element. The required `ReportDate` element identifies the testing laboratory report issue date using the XSD Date data type, YYYY-MM-DD.

4.2.4 Test Start Date Element. The required `TestStartDate` element identifies the test start date, using the XSD Date data type, YYYY-MM-DD.

4.2.5 Test End Date Element. The required `TestEndDate` element identifies the most recent date the data was collected for this report, using the XSD Date data type, YYYY-MM-DD.

4.2.6 Unique Identifier Element. The required `UniqueIdentifier` element contains a Universally Unique Identifier (UUID) as defined by RFC 4122 (see **Section 2.4**). Most scientific programming language libraries include functions that will automatically generate UUIDs.

A UUID is a unique 128-bit value expressed as a string with 32 hexadecimal digits in the format xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx. For example:

21EC2020-3AEA-4069-A2DD-08002B30309D

The intent of the UUID is to uniquely identify the document's measurement data. The document creator shall generate a new UUID whenever the document's measurement data is modified.

4.2.7 Comment Element. The optional `Comment` element provides arbitrary additional information relating to the tested and reported data.

Multiple `Comment` elements may be included.

4.2.8 Reference Element. The optional `Reference` element identifies one or more regulatory standards or other documents with which the luminaire or emitter complies, or the regulatory standards or other documents to which the laboratory measurements conform. This may, for example, be a document name or a web address.

Multiple `Reference` elements may be included.

4.2.9 More Information URI Element. The optional `MoreInfoURI` element provides a Uniform Resource Identifier (URI) address for further device information.

Multiple `MoreInfoURI` elements may be included.

4.3 Device Under Test Element

The required `DUT` element is the parent of the device under test section. `DUT` elements are listed in **Table 4-2**.

4.3.1 Manufacturer Element. The required `Manufacturer` element identifies the manufacturer of the DUT.

4.3.2 Catalog Number Element. The required `CatalogNumber` element identifies the manufacturer's product catalog number or similar identification.

4.3.3 Description Element. The required `Description` element contains a text description of the DUT, including whether it is an LED package, array, or module.

4.3.4 Seasoning Duration Element. The optional `SeasonDuration` element specifies the number of hours that the devices were seasoned prior to testing, as per Section 7.2 of ANSI/IES LM-80-21 (refer to **Section 2.2**).

Table 4-2. Device Under Test Element Fields

Element Description	Element Name	Data Type	Required/Optional	Document Section
Manufacturer	Manufacturer	xs:string	Required	4.3.1
Catalog Number	CatalogNumber	xs:string	Required	4.3.2
Description	Description	xs:string	Required	4.3.3
Seasoning Duration	SeasonDuration	xs:int	Optional	4.3.4
Sampling Method	SamplingMethod	xs:string	Optional	4.3.5

4.3.5 Sampling Method Element. The optional `SamplingMethod` element specifies the DUT sampling method, for example, random sampling or first production reel.

4.4 Test Equipment Element

The required `TestEquipment` element is the parent of the test equipment section. `TestEquipment` elements are listed in **Table 4-3**.

4.4.1 TMP Element. The required TMP element is the parent of the Temperature Measurement Point information.

4.4.1.1 Location Element. The required `Location` element describes the TMP position.

4.4.1.2 Image Element. The required `Image` element is a Base64 encoded image of the TMP location.

Table 4-3. Test Equipment Element Fields

Element Description	Element Name	Data Type	Required/Optional	Document Section
TMP Element	TMP	XML Element	Required	4.4.1
Location	Location	xs:string	Required	4.4.1.1
Image	Image	xs:string	Required	4.4.1.2
Life Test Equipment	LifeTestEquipment	XML Element	Required	4.4.2
System Type	SystemType	xs:string	Required	4.4.2.1
Description	Description	xs:string	Required	4.4.2.2
Manufacturer	Manufacturer	xs:string	Required	4.4.2.3
Model Number	ModelNumber	xs:string	Required	4.4.2.4
Calibration Date	CalibrationDate	xs:date	Required	4.4.2.5
Uncertainty	Uncertainty	xs:string	Optional	4.4.2.6
Radiometric Test Equipment	RadiometricTestEquipment	XML Element	Required	4.4.3
System Type	SystemType	xs:string	Required	4.4.3.1
Description	Description	xs:string	Required	4.4.3.2
Manufacturer	Manufacturer	xs:string	Required	4.4.3.3
Model Number	ModelNumber	xs:string	Required	4.4.3.4
Calibration Date	CalibrationDate	xs:date	Required	4.4.3.5
Uncertainty	Uncertainty	xs:string	Optional	4.4.3.6
Sphere Drift Correction	SphereDriftCorrection	xs:string	Optional	4.4.3.7
Temperature Correction	TemperatureCorrection	xs:string	Optional	4.4.3.8
Maximum Measurable Wavelength	MaxMeasureWavelength	xs:decimal	Optional	4.4.3.9
Minimum Measurable Wavelength	MinMeasureWavelength	xs:decimal	Optional	4.4.3.10
Peak Wavelength Method	PeakWavelengthMethod	xs:string	Optional	4.4.3.11
Electrical Drive Equipment	ElectricalDriveEquipment	XML Element	Required	4.4.4
System Type	SystemType	xs:string	Required	4.4.4.1
Description	Description	xs:string	Required	4.4.4.2
Manufacturer	Manufacturer	xs:string	Required	4.4.4.3
Model Number	ModelNumber	xs:string	Required	4.4.4.4
Calibration Date	CalibrationDate	xs:date	Required	4.4.4.5
Uncertainty	Uncertainty	xs:string	Optional	4.4.4.6

4.4.2 Life Test Equipment Element. The required `LifeTestEquipment` element is the parent of the elements related to the life test equipment.

4.4.2.1 SystemTypeElement. The required `SystemType` element specifies the system.

4.4.2.2 Description Element. The required `Description` element provides additional details about the life test system.

4.4.2.3 Manufacturer Element. The required `Manufacturer` element is the manufacturer of the life test equipment.

4.4.2.4 Model Number Element. The required `ModelNumber` element is the model number for the life test equipment.

4.4.2.5 Calibration Date Element. The required `CalibrationDate` element is the date of the last calibration of the equipment, using the XSD Date data type, YYYY-MM-DD.

4.4.2.6 Uncertainty Element. The optional `Uncertainty` element describes the uncertainty in the life test equipment.

4.4.3 Radiometric Test Equipment Element. The required `RadiometricTestEquipment` element is the parent of the elements related to the radiometric test equipment.

4.4.3.1 SystemTypeElement. The required `SystemType` element specifies the measurement system. Valid values include:

`IntegratingSphere`
`IntegratingHemisphere`
`Gonioradiometer`

Values other than these three may be used.

4.4.3.2 Description Element. The required `Description` element provides additional details about the measurement system.

4.4.3.3 Manufacturer Element. The required

`Manufacturer` element is the manufacturer of the radiometric test equipment.

4.4.3.4 Model Number Element. The required `ModelNumber` element is the model number for the radiometric test equipment.

4.4.3.5 Calibration Date Element. The required `CalibrationDate` element is the date of the last calibration of the equipment, using the XSD Date data type, YYYY-MM-DD.

4.4.3.6 Uncertainty Element. The optional `Uncertainty` element describes the uncertainty in the radiometric test equipment.

4.4.3.7 Sphere Drift Correction Element. The optional `SphereDriftCorrection` element describes any sphere drift corrections made to the measurement systems' responsivity as per Section 6.2 of normative reference ANSI/IES LM-80-21 (refer to **Section 2.2**). This element shall be included if any sphere drift correction was performed.

4.4.3.8 Temperature Correction Element. The optional `TemperatureCorrection` element describes any temperature corrections made to the measurement systems' responsivity. This element shall be included if any temperature correction was performed.

4.4.3.9 Maximum Measurable Wavelength Element. The optional `MaxMeasureWavelength` element contains the highest measurable wavelength based on the limitations of the radiometric test equipment.

4.4.3.10 Minimum Measurable Wavelength Element. The optional `MinMeasureWavelength` element contains the lowest measurable wavelength based on the limitations of the radiometric test equipment.

4.4.3.11 Peak Wavelength Method Element. The optional `PeakWavelengthMethod` element specifies the method of calculation for peak wavelength.

ANSI/IES LM-80-21 Section 3.10 indicates that for spectra which contain two peaks, the region for the peak of interest can be specified in a number of ways, such as "pump emission" or "phosphor emission," or as the peak within a specified wavelength band, such as "peak within 300 to 500 nm."

4.4.4 Electrical Drive Equipment Element. The required `ElectricalDriveEquipment` element is the parent of the elements related to the electrical drive equipment used for the radiometric test.

4.4.4.1 System Type Element. The required `SystemType` element specifies the measurement system. Valid values are:

`ConstCurrentDC`
`ConstVoltageDC`
`RegVoltageAC`
`PWM`

Values other than these four are *not* allowed, since these specific values are used in other places in the standard.

4.4.4.2 Description Element. The required `Description` element provides additional details about the electrical drive system.

Multiple `Description` elements may be included.

4.4.4.3 Manufacturer Element. The required `Manufacturer` element is the manufacturer of the electrical drive equipment.

4.4.4.4 Model Number Element. The required `ModelNumber` element is the model number for the electrical drive equipment.

4.4.4.5 Calibration Date Element. The required `CalibrationDate` element is the date of the last calibration of the equipment, using the XSD Date data type, YYYY-MM-DD.

4.4.4.6 Uncertainty Element. The optional `Uncertainty` element describes the uncertainty in the electrical drive equipment.

4.5 Measurement Set

The required `MeasurementSet` element is the parent of all measurement results. `MeasurementSet` elements are listed in **Table 4-4**.

Multiple `MeasurementSet` elements can be included. There will be one `MeasurementSet` element for each unique combination of test duration, electrical conditions, and thermal conditions.

4.5.1 Air Temperature Range Element. The optional `AirTempRange` element is the parent of the air temperature range data.

4.5.1.1 Minimum Temperature Element. The optional `MinTemp` element specifies the minimum air temperature, expressed in degrees Celsius.

Table 4-4. Measurement Set Element Fields

Element Description	Element Name	Data Type	Required / Optional	Document Section
Air Temperature Range	AirTempRange	XML Element	Optional	4.5.1
Minimum Temp	MinTemp	xs:decimal	Optional	4.5.1.1
Nominal Temp	NominalTemp	xs:decimal	Optional	4.5.1.2
Maximum Temp	MaxTemp	xs:decimal	Optional	4.5.1.3
TMP Temperature Range	TMPTempRange	XML Element	Required	4.5.2
Minimum Temp	MinTemp	xs:decimal	Required	4.5.2.1
Nominal Temp	NominalTemp	xs:decimal	Required	4.5.2.2
Maximum Temp	MaxTemp	xs:decimal	Required	4.5.2.3
Air Movement	AirMovement	xs:string	Required	4.5.3
Relative Humidity	RelativeHumidity	xs:string	Required	4.5.4
Elapsed Time	ElapsedTime	xs:integer	Required	4.5.5
Number of DUTs	NumberDUT	xs:integer	Required	4.5.6
DUTs	DUT	XML Element	Required	4.5.7

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4.5.1.2 Nominal Temperature Element. The optional `NominalTemp` element specifies the nominal air temperature, expressed in degrees Celsius.

4.5.1.3 Maximum Temperature Element. The optional `MaxTemp` element specifies the maximum air temperature, expressed in degrees Celsius.

4.5.2 TMP Temperature Range Element. The required `TMPTempRange` element is the parent of the TMP temperature range data.

4.5.2.1 Minimum Temperature Element. The required `MinTemp` element specifies the minimum TMP temperature, expressed in degrees Celsius.

4.5.2.2 Nominal Temperature Element. The required `NominalTemp` element specifies the nominal TMP temperature, expressed in degrees Celsius.

4.5.2.3 Maximum Temperature Element. The required `MaxTemp` element specifies the maximum TMP temperature, expressed in degrees Celsius.

4.5.3 Air Movement Element. The required

`AirMovement` element briefly characterizes the air movement near the DUT during the maintenance test (e.g., passive convection, active convection).

4.5.4 Relative Humidity Element. The required `RelativeHumidity` element specifies the chosen relative humidity, expressed in percent as an integer from 0 to 100, or if the standard humidity of less than or equal to 65% is used, this element shall be set to "std-65".

4.5.5 Elapsed Time Element. The required `ElapsedTime` element specifies the on time for this measurement set, expressed as an integer number of hours.

4.5.6 Number of DUTs Element. The required `NumberDUT` element specifies the total number of DUTs included in this test, including working and failed DUTs, as an integer number.

4.5.7 DUTs Element. The required DUT element is the parent of the DUT test data. There is one DUT element for each device under test. The DUT element items are detailed in **Table 4-5**.

Table 4-5: DUT Element Items

Element Description	Element Name	Data Type	Required/Optional	Document Section
Unique ID	UID	xs:string	Required	4.5.7.1
Failed	Failed	XML Element	Optional	4.5.7.2
Failure Mode	FailureMode	xs:string	Required	4.5.7.2.1
Time of Failure	TimeOfFailure	xs:decimal	Required	4.5.7.2.2
Voltage	Voltage	xs:decimal	Required	4.5.7.3
Current	Current	xs:decimal	Required	4.5.7.4
Luminous Flux	LuminousFlux	xs:decimal	Optional / Choice	4.5.7.5
Photon Flux	PhotonFlux	xs:decimal	Optional / Choice	4.5.7.6
Radiant Flux	RadiantFlux	xs:decimal	Optional / Choice	4.5.7.7
Chromaticity 1931 x	Chromaticity_1931_x	xs:decimal	Optional	4.5.7.8
Chromaticity 1931 y	Chromaticity_1931_y	xs:decimal	Optional	4.5.7.9
Chromaticity 1976 u'	Chromaticity_1976_up	xs:decimal	Required	4.5.7.10
Chromaticity 1976 v'	Chromaticity_1976_vp	xs:decimal	Required	4.5.7.11
Peak Wavelength	PeakWavelength	xs:decimal	Optional	4.5.7.12
Dominant Wavelength	DominantWavelength	xs:decimal	Optional	4.5.7.13
Centroid Wavelength	CentroidWavelength	xs:decimal	Optional	4.5.7.14
Spectral Power Distribution	SpectralPowerDistribution	XML Element	Optional	4.5.7.15

4.5.7.1 Unique ID Element. The required `UID` element is a unique ID assigned to each device under test to allow tracking of a specific device through multiple `MeasurementSet` elements.

4.5.7.2 Failed Element. The optional `Failed` element is the parent of the failed DUT data. This element shall be included if this DUT failed during testing based on normative reference ANSI/IES LM-80-21 (see **Section 2.2**).

4.5.7.2.1 Failure Mode Element. The required `FailureMode` element specifies the DUT failure mode. Typical failure modes are mechanical, electrical, and optical.

4.5.7.2.2 Time of Failure Element. The required `TimeOfFailure` element specifies the DUT failure time, expressed as the integer number of hours since test start.

4.5.7.3 Voltage Element. The required `Voltage` element contains the drive voltage for this DUT if it is voltage driven, or the measured forward voltage if current driven. If the device is no longer producing detectable radiometric output, this shall be set to 0.

4.5.7.4 Current Element. The required `Current` element contains the drive current for this DUT if current driven, or the measured forward current if voltage driven. If the Electrical Drive Equipment System Type in **Section 4.4.4.1** is PWM, the `Current` element may be omitted. If the device is no longer producing detectable radiometric output, this element shall be set to 0.

4.5.7.5 Luminous Flux Element. The optional `LuminousFlux` element contains flux output of the DUT, expressed in lumens. While `LuminousFlux` is not required, at least one of the three flux elements is required, and all may be included. If the device is no longer producing detectable radiometric output, this element shall be set to 0 if it was previously being reported.

4.5.7.6 Photon Flux Element. The optional `PhotonFlux` element contains flux output of the DUT, expressed in photon flux with the units of $\mu\text{mol/s}$. While `PhotonFlux`

is not required, at least one of the three flux elements is required, and all may be included. If the device is no longer producing detectable radiometric output, this element shall be set to 0 if it was previously being reported.

4.5.7.7 Radiant Flux Element. The optional `RadiantFlux` element contains flux output of the DUT, expressed in radiant flux with the units of radiant watts. While `RadiantFlux` is not required, at least one of the three flux elements is required, and all may be included. If the device is no longer producing detectable radiometric output, this element shall be set to 0 if it was previously being reported.

4.5.7.8 Chromaticity 1931 x Element. The optional `Chromaticity_1931_x` element is the x coordinate of the chromaticity in the 1931 CIE color space. If the device is no longer producing detectable radiometric output, this element shall be set to 0 if it was previously being reported.

4.5.7.9 Chromaticity 1931 y Element. The optional `Chromaticity_1931_y` element is the y coordinate of the chromaticity in the 1931 CIE color space. If the device is no longer producing detectable radiometric output, this element shall be set to 0 if it was previously being reported.

4.5.7.10 Chromaticity 1976 u' Element. The required `Chromaticity_1976_up` element is the u' coordinate of the chromaticity in the 1976 CIE color space. If the device is no longer producing detectable radiometric output, this element shall be set to 0.

4.5.7.11 Chromaticity 1976 v' Element. The required `Chromaticity_1976_vp` element is the v' coordinate of the chromaticity in the 1976 CIE color space. If the device is no longer producing detectable radiometric output, this element shall be set to 0.

4.5.7.12 Peak Wavelength Element. The optional `PeakWavelength` element contains the peak wavelength for this DUT. If the device is no longer producing detectable radiometric output, this element shall be set to 0 if it was previously being reported.

4.5.7.13 Dominant Wavelength Element. The optional `DominantWavelength` element contains the dominant wavelength for this DUT. If the device is no longer producing detectable radiometric output, this element shall be set to 0 if it was previously being reported.

4.5.7.14 Centroid Wavelength Element. The optional `CentroidWavelength` element contains the centroid wavelength for this DUT. If the device is no longer producing detectable radiometric output, this element shall be set to 0 if it was previously being reported.

4.5.7.15 Spectral Power Distribution Element. The optional `SpectralPowerDistribution` element contains the spectral power distribution for this DUT, formatted according to ANSI/IES TM-27-20 (see **Section 2.3**). If the device is no longer producing detectable radiometric output, this element shall be removed.

Inclusion of the `SpectralPowerDistribution` element makes it possible to convert between and compute any flux value (luminous, radiant, photon).

4.6 Custom Data Element

The optional `CustomData` element is the parent of custom data. Elements are detailed in **Table 4-6**.

The intent of the `CustomData` element is to provide a means of including custom data within the luminaire optical-data document. For example, a company or

government agency may require additional information that cannot be represented within the base XML schema (see **Section 5**).

Multiple `CustomData` elements are allowed.

4.6.1 Name Element. The required `Name` element specifies the name describing the custom data.

4.6.2 Unique Identifier Element. The required `UniqueIdentifier` element contains a Universally Unique Identifier (UUID), as defined by RFC 4122 (see **Section 2.4**). Most scientific programming language libraries include functions that will automatically generate UUIDs.

A UUID is a unique 128-bit value expressed as a string with 32 hexadecimal digits in the format xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx. For example:

E1EC2128-3AEB-1069-A24E-98712B54306F

The purpose of this UUID is to prevent name collisions between different instances of the `Name` element.

4.6.3 Any Data Element. The base XML schema (see **Section 5**) allows an arbitrary sequence of XML elements within the `CustomData` element using an `<any>` declaration with a `##any` namespace attribute to instruct a validating parser to ignore the element sequence for validation purposes. This is an optional element.

Table 4-6. Custom Data Element Fields

Element Description	Element Name	Data Type	Required/Optional	Document Section
Name	Name	xs:string	Required	4.6.1
Unique Identifier	UniqueIdentifier	xs:string	Required	4.6.2
Any Data	[Not applicable]	XML Element	Optional	4.6.3

5.0 XML Schema

The XML schema follows:

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="IESLM80">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="Version" type="xs:string" fixed="1.0"/>
        <xs:element name="Admin">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="TestAgency" type="xs:string"/>
              <xs:element name="ReportNumber" type="xs:string"/>
              <xs:element name="ReportDate" type="xs:date"/>
              <xs:element name="TestStartDate" type="xs:date"/>
              <xs:element name="TestEndDate" type="xs:date"/>
              <xs:element name="UniqueIdentifier" type="xs:string"/>
              <xs:element name="Comment" type="xs:string" minOccurs="0"
                maxOccurs="unbounded"/>
              <xs:element name="Reference" type="xs:string" minOccurs="0"
                maxOccurs="unbounded"/>
              <xs:element name="MoreInfoURI" type="xs:string" minOccurs="0"
                maxOccurs="unbounded"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
        <xs:element name="DUT">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="Manufacturer" type="xs:string"/>
              <xs:element name="CatalogNumber" type="xs:string"/>
              <xs:element name="Description" type="xs:string"/>
              <xs:element name="SeasonDuration" type="xs:int" minOccurs="0"/>
              <xs:element name="SamplingMethod" type="xs:string" minOccurs="0"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
        <xs:element name="TestEquipment">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="TMP">
                <xs:complexType>
                  <xs:sequence>
```

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```
        <xs:element name="Location" type="xs:string"/>
        <xs:element name="Image" type="xs:string"/>
    </xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="LifeTestEquipment">
    <xs:complexType>
        <xs:sequence>
            <xs:element name="SystemType" type="xs:string"/>
            <xs:element name="Description" type="xs:string"/>
            <xs:element name="Manufacturer" type="xs:string"/>
            <xs:element name="ModelNumber" type="xs:string"/>
            <xs:element name="CalibrationDate" type="xs:date"/>
            <xs:element name="Uncertainty" type="xs:string" minOccurs="0"/>
        </xs:sequence>
    </xs:complexType>
</xs:element>
<xs:element name="RadiometricTestEquipment">
    <xs:complexType>
        <xs:sequence>
            <xs:element name="SystemType" type="xs:string"/>
            <xs:element name="Description" type="xs:string"/>
            <xs:element name="Manufacturer" type="xs:string"/>
            <xs:element name="ModelNumber" type="xs:string"/>
            <xs:element name="CalibrationDate" type="xs:date"/>
            <xs:element name="Uncertainty" type="xs:string" minOccurs="0">
                <xs:element name="SphereDriftCorrection" type="xs:string"
                    minOccurs="0"/>
                <xs:element name="TemperatureCorrection" type="xs:string"
                    minOccurs="0"/>
                <xs:element name="MaxMeasureWavelength" type="xs:decimal"
                    minOccurs="0"/>
                <xs:element name="MinMeasureWavelength" type="xs:decimal"
                    minOccurs="0"/>
                <xs:element name="PeakWavelengthMethod" type="xs:string"
                    minOccurs="0"/>
            </xs:sequence>
        </xs:complexType>
</xs:element>
<xs:element name="ElectricalDriveEquipment">
    <xs:complexType>
        <xs:sequence>
            <xs:element name="SystemType">
                <xs:simpleType>
                    <xs:restriction base="xs:string">
                        <xs:enumeration value="ConstCurrentDC"/>
                        <xs:enumeration value="ConstVoltageDC"/>
                    </xs:restriction>
                </xs:simpleType>
            </xs:element>
        </xs:sequence>
    </xs:complexType>
</xs:element>
```

```

        <xs:enumeration value="RegVoltageAC"/>
        <xs:enumeration value="PWM"/>
    </xs:restriction>
</xs:simpleType>
</xs:element>
<xs:element name="Description" type="xs:string"/>
<xs:element name="Manufacturer" type="xs:string"/>
<xs:element name="ModelNumber" type="xs:string"/>
<xs:element name="CalibrationDate" type="xs:date"/>
<xs:element name="Uncertainty" type="xs:string" minOccurs="0"/>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="MeasurementSet" maxOccurs="unbounded">
    <xs:complexType>
        <xs:sequence>
            <xs:element name="AirTempRange" minOccurs="0">
                <xs:complexType>
                    <xs:sequence>
                        <xs:element name="MinTemp" type="xs:decimal"/>
                        <xs:element name="NominalTemp" type="xs:decimal"/>
                        <xs:element name="MaxTemp" type="xs:decimal"/>
                    </xs:sequence>
                </xs:complexType>
            </xs:element>
            <xs:element name="TMPTempRange">
                <xs:complexType>
                    <xs:sequence>
                        <xs:element name="MinTemp" type="xs:decimal"/>
                        <xs:element name="NominalTemp" type="xs:decimal"/>
                        <xs:element name="MaxTemp" type="xs:decimal"/>
                    </xs:sequence>
                </xs:complexType>
            </xs:element>
            <xs:element name="AirMovement" type="xs:string"/>
            <xs:element name="RelativeHumidity" type="xs:string"/>
            <xs:element name="ElapsedTime" type="xs:int"/>
            <xs:element name="NumberDUT" type="xs:int"/>
            <xs:element name="DUTs">
                <xs:complexType>
                    <xs:sequence>
                        <xs:element name="UID" type="xs:string"/>
                        <xs:element name="Failed" minOccurs="0">
                            <xs:complexType>

```

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```
<xs:sequence>
  <xs:element name="FailureMode" type="xs:string"/>
  <xs:element name="TimeOfFailure" type="xs:int"/>
</xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="Voltage" type="xs:decimal"/>
<xs:element name="Current" type="xs:decimal" minOccurs="0"/>
<xs:element name="LuminousFlux" type="xs:decimal" minOccurs="0"/>
<xs:element name="PhotonFlux" type="xs:decimal" minOccurs="0"/>
<xs:element name="RadiantFlux" type="xs:decimal" minOccurs="0"/>
  <xs:element name="Chromaticity_1931_x" type="xs:decimal"
    minOccurs="0"/>
  <xs:element name="Chromaticity_1931_y" type="xs:decimal"
    minOccurs="0"/>
  <xs:element name="Chromaticity_1976_up" type="xs:decimal"/>
  <xs:element name="Chromaticity_1976_vp" type="xs:decimal"/>
<xs:element name="PeakWavelength" type="xs:decimal" minOccurs="0"/>
  <xs:element name="DominantWavelength" type="xs:decimal"
    minOccurs="0"/>
  <xs:element name="CentroidWavelength" type="xs:decimal"
    minOccurs="0"/>
<xs:element name="SpectralPowerDistribution" minOccurs="0">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="Ident" type="xs:int"/>
      <xs:element name="WM" maxOccurs="unbounded">
        <xs:complexType>
          <xs:simpleContent>
            <xs:extension base="xs:decimal">
              <xs:attribute name="w" type="xs:int"/>
            </xs:extension>
          </xs:simpleContent>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="CustomData" minOccurs="0" maxOccurs="unbounded">
  <xs:complexType>
    <xs:sequence>
```

```

        <xs:element name="Name" type="xs:string"/>
        <xs:element name="UniqueIdentifier" type="xs:string"/>
            <xs:any namespace="##any" processContents="skip" minOccurs="0"
                maxOccurs="unbounded"/>
        </xs:sequence>
    </xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:schema>

```

6.0 JSON Document Format

JavaScript Object Notation (JSON) is a minimalist data-interchange format that is widely used for internet client-server communications. Unlike the rich semantics of XML with its many data types, JSON allows only name-value pairs and ordered lists.

The advantage of the XML document format in representing measurement data is that the documents can be validated using the associated schema (see **Section 5**); the advantage of the JSON document format is that its ordered lists can be used to compactly represent data arrays. In particular, JSON documents tend to be an order of magnitude smaller than their equivalent XML counterparts.

6.1 XML to JSON Mapping

Mapping XML elements to JSON name-value pairs is straightforward. For example, this XML complex element:

```

<DUT>
  <Manufacturer>Academy Lighting</Manufacturer>
  <CatalogNumber>ALLM 45-003-W3500</CatalogNumber>
  <Description>3500K White LED array</Description>
  <NumberDUT>10</NumberDUT>
</DUT>

```

becomes this in JSON:

```

"DUT": {
  "Manufacturer": "Academy Lighting",
  "CatalogNumber": "ALLM 45-003-W3500",
  "Description": "3500K White LED array",
  "NumberDUT": 10
}

```

Where multiple XML elements are allowed, these can be represented in JSON as arrays. For example, this:

```

<Comment>Sample XML file</Comment>
<Comment>Fictitious manufacturer</Comment>

```

becomes this:

```
"Comment": [  
  "Sample JSON file",  
  "Fictitious manufacturer"  
]
```

6.2 JSON Schema

The JSON schema follows:

```
{  
  "$schema": "http://json-schema.org/draft/2019-09/schema#",  
  "type": "object",  
  "properties": {  
    "Version": {  
      "type": "number"  
    },  
    "Admin": {  
      "type": "object",  
      "properties": {  
        "TestAgency": {  
          "type": "string"  
        },  
        "ReportNumber": {  
          "type": "string"  
        },  
        "ReportDate": {  
          "type": "string",  
          "pattern": "^(\\[0-9\\]{4}-[0-9]{2}-[0-9]{2})$|UNKNOWN"  
        },  
        "TestStartDate": {  
          "type": "string",  
          "pattern": "^(\\[0-9\\]{4}-[0-9]{2}-[0-9]{2})$|UNKNOWN"  
        },  
        "TestEndDate": {  
          "type": "string",  
          "pattern": "^(\\[0-9\\]{4}-[0-9]{2}-[0-9]{2})$|UNKNOWN"  
        },  
        "UniqueIdentifier": {  
          "type": "string"  
        },  
        "Comments": {  
          "type": "array",  
          "items": {  
            "type": "string"  
          }  
        },  
        "Reference": {  
          "type": "array",
```

```

        "items": {
            "type": "string"
        }
    },
    "MoreInfoURI": {
        "type": "string"
    }
},
"required": [
    "TestAgency",
    "ReportNumber",
    "ReportDate",
    "TestStartDate",
    "TestEndDate",
    "UniqueIdentifier"
]
},
"DUT": {
    "type": "object",
    "properties": {
        "Manufacturer": {
            "type": "string"
        },
        "CatalogNumber": {
            "type": "string"
        },
        "Description": {
            "type": "string"
        },
        "SeasonDuration": {
            "type": "number"
        },
        "SamplingMethod": {
            "type": "string"
        }
    },
    "required": [
        "Manufacturer",
        "CatalogNumber",
        "Description"
    ]
},
"TestEquipment": {
    "type": "object",
    "properties": {
        "TMP": {
            "type": "object",

```

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```
"properties": {
  "Location": {
    "type": "string"
  },
  "Image": {
    "type": "string"
  }
},
"required": [
  "Location",
  "Image"
]
},
"RadiometricTestEquipment": {
  "type": "object",
  "properties": {
    "SystemType": {
      "type": "string"
    },
    "Description": {
      "type": "string"
    },
    "Manufacturer": {
      "type": "string"
    },
    "ModelNumber": {
      "type": "string"
    },
    "CalibrationDate": {
      "type": "string"
    },
    "Uncertainty": {
      "type": "string"
    },
    "SphereDriftCorrection": {
      "type": "string"
    },
    "TemperatureCorrection": {
      "type": "string"
    },
    "MaxMeasureWavelength": {
      "type": "number"
    },
    "MinMeasureWavelength": {
      "type": "number"
    },
    "PeakWavelengthMethod": {
```

```

        "type": "string"
    }
},
"required": [
    "SystemType",
    "Description",
    "Manufacturer",
    "ModelNumber",
    "CalibrationDate"
]
},
"ElectricalDriveEquipment": {
    "type": "object",
    "properties": {
        "SystemType": {
            "type": "string",
            "enum": [
                "ConstCurrentDC",
                "ConstVoltageDC",
                "RegVoltageAC",
                "PWM"
            ]
        },
        "Description": {
            "type": "string"
        },
        "Manufacturer": {
            "type": "string"
        },
        "ModelNumber": {
            "type": "string"
        },
        "CalibrationDate": {
            "type": "string"
        },
        "Uncertainty": {
            "type": "string"
        }
    },
    "required": [
        "SystemType",
        "Description",
        "Manufacturer",
        "ModelNumber",
        "CalibrationDate"
    ]
},

```

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```
    "required": [
      "TMP",
      "RadiometricTestEquipment",
      "ElectricalDriveEquipment"
    ]
  },
  "MeasurementSet": {
    "type": "array",
    "items": {
      "AirTempRange": {
        "type": "object",
        "properties": {
          "MinTemp": {
            "type": "number"
          },
          "NominalTemp": {
            "type": "number"
          },
          "MaxTemp": {
            "type": "number"
          }
        }
      },
      "required": [
        "MinTemp",
        "NominalTemp",
        "MaxTemp"
      ]
    },
    "TMPTempRange": {
      "type": "object",
      "properties": {
        "MinTemp": {
          "type": "number"
        },
        "NominalTemp": {
          "type": "number"
        },
        "MaxTemp": {
          "type": "number"
        }
      },
      "required": [
        "MinTemp",
        "NominalTemp",
        "MaxTemp"
      ]
    }
  ]
}
```

```

    },
    "AirMovement": {
      "type": "number"
    },
    "RelativeHumidity": {
      "type": "string"
    },
    "ElapsedTime": {
      "type": "number"
    },
    "NumberDUT": {
      "type": "number"
    },
    "DUTs": {
      "type": "array",
      "items": {
        "UID": {
          "type": "string"
        },
        "Voltage": {
          "type": "number"
        },
        "Current": {
          "type": "number"
        },
        "LuminousFlux": {
          "type": "number"
        },
        "PhotonFlux": {
          "type": "number"
        },
        "RadiantFlux": {
          "type": "number"
        },
        "Chromaticity_1931_x": {
          "type": "number"
        },
        "Chromaticity_1931_y": {
          "type": "number"
        },
        "Chromaticity_1976_up": {
          "type": "number"
        },
        "Chromaticity_1976_vp": {
          "type": "number"
        },
        "PeakWavelength": {

```

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```
        "type": "number"
      },
      "DominantWavelength": {
        "type": "number"
      },
      "CentroidWavelength": {
        "type": "number"
      }
    },
    "required": [
      "UID",
      "Voltage",
      "Chromaticity _ 1976 _ up",
      "Chromaticity _ 1976 _ vp"
    ]
  },
  "required": [
    "TMPTempRange",
    "AirMovement",
    "RelativeHumidity",
    "ElapsedTime",
    "NumberDUT",
    "DUTs"
  ]
}
},
"required": [
  "Version",
  "Admin",
  "DUT",
  "TestConditions",
  "MeasurementSet"
]
}
}
```

Annex A – Sample Documents

Note: This annex is not part of ANSI/IES TM-41-24. It is included for informational purposes only.

Sample XML and JSON documents are provided in **Sections A.1** and **A.2**, respectively.

A.1 Sample XML Document

```
<?xml version="1.0" encoding="UTF-8"?>
<IESLM80>
  <Version>1.0</Version>
  <Admin>
    <TestAgency>Analytical Laboratory A</TestAgency>
    <ReportNumber>ALA-220415-01</ReportNumber>
    <ReportDate>2022-04-15</ReportDate>
    <TestStartDate>2021-01-01</TestStartDate>
    <TestEndDate>2021-12-31</TestEndDate>
    <UniqueIdentifier>21EC2020-3AEA-A2DD-08002B30309E</UniqueIdentifier>
    <Comment>Sample XML file</Comment>
  </Admin>
  <DUT>
    <Manufacturer>LED Company A</Manufacturer>
    <CatalogNumber>ALLM 45-003-W3500</CatalogNumber>
    <Description>3500K White LED</Description>
  </DUT>
  <TestEquipment>
    <TMP>
      <Location>Tc near positive LED connection</Location>
      <Image>...</Image>
    </TMP>
    <LifeTestEquipment>
      <SystemType>Chamber</SystemType>
      <Description>Thermal Chamber</Description>
      <Manufacturer>LifeTest Manufacturer B</Manufacturer>
      <ModelNumber>LT-C</ModelNumber>
      <CalibrationDate>2020-01-01</CalibrationDate>
      <Uncertainty>temp value to +/- 2deg</Uncertainty>
    </LifeTestEquipment>
    <RadiometricTestEquipment>
      <SystemType>IntegratingSphere</SystemType>
      <Description>2m diameter integrating sphere</Description>
      <Manufacturer>Sphere Manufacturer B</Manufacturer>
      <ModelNumber>IS-B</ModelNumber>
      <CalibrationDate>2020-01-01</CalibrationDate>
      <Uncertainty>lumen value to +/- 1.2%</Uncertainty>
    </RadiometricTestEquipment>
```

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```
<ElectricalDriveEquipment>
  <SystemType>ConstCurrentDC</SystemType>
  <Description>DC power supply</Description>
  <Manufacturer>Power Supply Manufacturer B</Manufacturer>
  <ModelNumber>PS-B</ModelNumber>
  <CalibrationDate>2020-02-02</CalibrationDate>
  <Uncertainty>voltage value to +/- 2%, current value to +/- 1%</Uncertainty>
</ElectricalDriveEquipment>
</TestEquipment>
<MeasurementSet>
  <TMPTempRange>
    <MinTemp>23.2</MinTemp>
    <NominalTemp>25</NominalTemp>
    <MaxTemp>27.8</MaxTemp>
  </TMPTempRange>
  <AirMovement>passive convection</AirMovement>
  <RelativeHumidity>std-65</RelativeHumidity>
  <ElapsedTime>3000</ElapsedTime>
  <NumberDUT>1038</NumberDUT>
  <DUTs>
    <UID>1037</UID>
    <Voltage>3.22</Voltage>
    <Current>0.350</Current>
    <LuminousFlux>118</LuminousFlux>
    <PhotonFlux>20</PhotonFlux>
    <Chromaticity _ 1976 _ up>0.241</Chromaticity _ 1976 _ up>
    <Chromaticity _ 1976 _ vp>0.341</Chromaticity _ 1976 _ vp>
  </DUTs>
  <DUTs>
    <UID>138</UID>
    <Voltage>3.24</Voltage>
    <Current>0.350</Current>
    <LuminousFlux>123</LuminousFlux>
    <PhotonFlux>25</PhotonFlux>
    <Chromaticity _ 1976 _ up>0.248</Chromaticity _ 1976 _ up>
    <Chromaticity _ 1976 _ vp>0.339</Chromaticity _ 1976 _ vp>
  </DUTs>
  <DUTs>
    <UID>1022</UID>
    <Failed>
      <FailureMode>mechanical</FailureMode>
      <TimeOfFailure>1500</TimeOfFailure>
    </Failed>
    <Voltage>3.25</Voltage>
    <Current>0.350</Current>
    <LuminousFlux>130</LuminousFlux>
    <PhotonFlux>30</PhotonFlux>
```

```

    <Chromaticity _ 1976 _ up>0.248</Chromaticity _ 1976 _ up>
    <Chromaticity _ 1976 _ vp>0.339</Chromaticity _ 1976 _ vp>
  </DUTs>
  <DUTs>
    <UID>85</UID>
    <Failed>
      <FailureMode>optical</FailureMode>
      <TimeOfFailure>1500</TimeOfFailure>
    </Failed>
    <Voltage>0</Voltage>
    <Current>0</Current>
    <LuminousFlux>0</LuminousFlux>
    <PhotonFlux>0</PhotonFlux>
    <Chromaticity _ 1976 _ up>0</Chromaticity _ 1976 _ up>
    <Chromaticity _ 1976 _ vp>0</Chromaticity _ 1976 _ vp>
  </DUTs>
</MeasurementSet>
</IESLM80>

```

A.2 Sample JSON Document

```

{
  "Version": 1.0,
  "Admin": {
    "TestAgency": "Analytical Laboratory A",
    "ReportNumber": "ALA-220415-01",
    "ReportDate": "2022-04-15",
    "TestStartDate": "2021-01-01",
    "TestEndDate": "2021-12-31",
    "UniqueIdentifier": "21EC2020-3AEA-A2DD-08002B30309E",
    "Comment": [
      "Sample JSON file"
    ]
  },
  "DUT": {
    "Manufacturer": "LED Company A",
    "CatalogNumber": "ALLM 45-003-W3500",
    "Description": "3500K White LED"
  },
  "TestEquipment": {
    "TMP": {
      "Location": "Tc near positive LED connection",
      "Image": "..."
    },
    "LifeTestEquipment": {
      "SystemType": "Chamber",
      "Description": "Thermal Chamber",
      "Manufacturer": "LifeTest Manufacturer B",

```

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```
"ModelNumber": "LT-C",
"CalibrationDate": "2020-01-01",
"Uncertainty": "temp value to +/- 2deg"
},
"RadiometricTestEquipment": {
  "SystemType": "IntegratingSphere",
  "Description": "2m diameter integrating sphere",
  "Manufacturer": "Sphere Manufacturer B",
  "ModelNumber": "IS-B",
  "CalibrationDate": "2020-01-01",
  "Uncertainty": "lumen value to +/- 1.2%"
},
"ElectricalDriveEquipment": {
  "SystemType": "ConstCurrentDC",
  "Description": "DC power supply",
  "Manufacturer": "Power Supply Manufacturer B",
  "ModelNumber": "PS-B",
  "CalibrationDate": "2020-02-02",
  "Uncertainty": "voltage value to +/- 2%, current value to +/- 1%"
}
},
"MeasurementSet": {
  "TMPTempRange": {
    "MinTemp": 23.2,
    "NominalTemp": 25.0,
    "MaxTemp": 27.8
  },
  "AirMovement": "passive convection",
  "RelativeHumidity": "std-65",
  "ElapsedTime": 3000,
  "NumberDUT": 1038,
  "DUTs": [
    {
      "UID": 1037,
      "Voltage": 3.22,
      "Current": 0.35,
      "LuminousFlux": 118,
      "PhotonFlux": 20,
      "Chromaticity_1976_up": 0.241,
      "Chromaticity_1976_vp": 0.341
    },
    {
      "UID": 138,
      "Voltage": 3.24,
      "Current": 0.35,
      "LuminousFlux": 123,
      "PhotonFlux": 25,
```

```

        "Chromaticity_1976_up": 0.248,
        "Chromaticity_1976_vp": 0.339
    },
    {
        "UID": 1022,
        "Failed": {
            "FailureMode": "mechanical",
            "TimeOfFailure": 1500
        },
        "Voltage": 3.25,
        "Current": 0.35,
        "LuminousFlux": 123,
        "PhotonFlux": 25,
        "Chromaticity_1976_up": 0.248,
        "Chromaticity_1976_vp": 0.339
    },
    {
        "UID": 85,
        "Failed": {
            "FailureMode": "optical",
            "TimeOfFailure": 1500
        },
        "Voltage": 0,
        "Current": 0,
        "LuminousFlux": 0,
        "PhotonFlux": 0,
        "Chromaticity_1976_up": 0,
        "Chromaticity_1976_vp": 0
    }
]
}
}

```

Annex B

Document Structure Overview

Note: This annex is not part of ANSI/IES TM-41-24. It is included for informational purposes only.

The following is a visual overview of the document structure.

	Data Type	Required	Multiple Allowed	Notes
Admin	XML Element	Required	No	
TestAgency	string	Required	No	
ReportNumber	string	Required	No	
ReportDate	date	Required	No	
TestStartDate	date	Required	No	
TestEndDate	date	Required	No	
UniquelIdentifier	string	Required	No	
Comment	string	Optional	Yes	
Reference	string	Optional	Yes	
More Information URI	URI	Optional	Yes	
DUT	XML Element	Required	No	
Manufacturer	string	Required	No	
CatalogNumber	string	Required	No	
Description	string	Required	No	
SeasonDuration	integer	Optional	No	
SamplingMethod	string	Optional	No	
TestEquipment	XML Element	Required	No	
TMP	XML Element	Required	No	
Location	string	Required	No	
Image	string	Required	No	
LifeTestEquipment	XML Element	Required	No	
SystemType	string	Required	No	
Description	string	Required	No	
Manufacturer	string	Required	No	
Model Number	string	Required	No	
Calibration Date	date	Required	No	
Uncertainty	string	Optional	No	

continued on next page

	Data Type	Required	Multiple Allowed	Notes
RadiometricTestEquipment	XML Element	Required	No	-Typical Options: IntegratingSphere, IntegratingHemisphere, or Gonioradiometer (these will be suggested as typical options, but the field is freeform text)
SystemType	string	Required	No	
Description	string	Required	No	
Manufacturer	string	Required	No	
Model Number	string	Required	No	
Calibration Date	date	Required	No	
Uncertainty	string	Optional	No	
SphereDriftCorrection	string	Optional	No	
TemperatureCorrection	string	Optional	No	
MaxMeasureWavelength	decimal	Optional	No	
MinMeasureWavelength	decimal	Optional	No	
PeakWavelengthMethod	string	Optional	No	
ElectricalDriveEquipment	XML Element	Required	No	<- Valid options are ConstCurrentDC, ConstVoltageDC, RegVoltageAC, PWM
SystemType	string	Required	No	
Description	string	Required	No	
Manufacturer	string	Required	No	
Model Number	string	Required	No	
Calibration Date	date	Required	No	
Uncertainty	string	Optional	No	
MeasurementSet	XML Element	Required	Yes	
AirTempRange	XML Element	Optional	No	
MinTemp	decimal	Optional	No	
NominalTemp	decimal	Optional	No	
MinMaxTemp	decimal	Optional	No	
TMPTempRange	XML Element	Required	No	
MinTemp	decimal	Required	No	
NominalTemp	decimal	Required	No	
MinMaxTemp	decimal	Required	No	

continued on next page

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	Data Type	Required	Multiple Allowed	Notes
AirMovement	string	Required	No	
RelativeHumidity	string	Required	No	
ElapsedTime	integer	Required	No	
NumberDUT	integer	Required	No	
DUT	XML Element	Required	Yes	<- one DUT element for every device
UID	string	Required	No	
Failed	XML Element	Optional	No	
FailureMode	string	Required	No	
TimeOfFailure	integer	Required	No	
Voltage	decimal	Required	No	
Current	decimal	Optional/ Conditional Required	No	
LuminousFlux	decimal	Optional/ Choice	No	<- at least one of the three flux options is required
PhotonFlux	decimal	Optional/ Choice	No	
RadiantFlux	decimal	Optional/ Choice	No	
Chromaticity_1931_x	decimal	Optional	No	
Chromaticity_1931_y	decimal	Optional	No	
Chromaticity_1976_up	decimal	Required	No	
Chromaticity_1976_vp	decimal	Required	No	
PeakWavelength	decimal	Optional	No	
DominantWavelength	decimal	Optional	No	
CentroidWavelength	decimal	Optional	No	
SpectralPowerDistribution	XML Element	Optional	No	<- element following the TM-27 format
CustomData	XML Element	Optional	Yes	
Name	string	Required	No	
UniquelIdentifier	string	Required	No	
Any Data	XML Element	Optional	No	

Informative References

- 1 Illuminating Engineering Society. ANSI/IES TM-21-21, Technical Memorandum: Projecting Long-Term Luminous, Photon, and Radiant Flux Maintenance of LED Light Sources. New York: IES; 2021.

Process for Change to an ANSI/IES Standard Under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard Under Continuous Maintenance

NOTE:

Use a separate form for each comment. Submit to the Director of Standards, IES, 85 Broad St, FL 17, New York NY 10004.

Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
Address: _____
City: _____ State: _____ Zip: _____ Country: _____
Telephone: _____
Fax: _____
E-mail: _____

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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